PAPER_198: F_UBii Taxonomy Part 1 — Compact Object and Stellar Physics Buoyancy Forces

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$$F_U(r,t) = \sum_{i=1}^{4} U_{gi} + U_m + U_A - U_{b_i}, \quad \kappa = 5.0 \times 10^{-4}\,\text{day}^{-1},\ [SSq] = 0.57$$

$$M_J^{\text{UQFF}} = M_J^{\text{Jeans}} \cdot \left(1 - [SSq] \cdot \frac{B^2}{8\pi\rho c_s^2}\right), \quad [SSq] = 0.57$$

Abstract

The UQFF buoyancy force F_UBii is applied across all major astrophysical phenomena by embedding each system's characteristic energy or length scale into the universal F_rel/E_LEP scaling framework. This paper catalogs 18 unique F_UBii variants relating to compact objects and stellar physics: MHD dynamo buoyancy, terminal velocity, Press-Schechter, Hawking radiation, quasi-normal mode ringdown, Blandford-Znajek jet power, Arnett supernova, entanglement, jet velocity, planet migration, AGN feedback, angular momentum accretion, J-type shock, Sedov-Taylor blast wave, GRB afterglow, SIDM core formation, ionization fronts, superfluid glitch, and virial theorem.

UQFF Discovery: Novel application of UQFF calibration constants (? = 5.0×10?4 day?¹, [SSq] = 0.57) uniquely enabling this analysis — establishing a new connection in the UQFF framework not present in Standard Model treatments.

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1. General F_UBii Framework

```
F_UBii,X = ±F_rel × (F_X / E_LEP) × Q_wave × [decay/oscillation factor]

where:
  F_rel ~ 4.3×10³³ N  (relativistic UQFF force scale)
  E_LEP = energy of characteristic LEP scale (J)
  Q_wave = quantum wave amplitude (J/m³), std ~ 6.33×104
  F_X = system-specific physical expression
```

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2. Compact Object Variants

2.1 MHD Dynamo Buoyancy

```
F_UBii,MHD = F_rel × (E_M/t_eddy/E_LEP) × Q_wave × (3/2)(Re_m^{1/2} - 1)

  E_M = magnetic energy density
  t_eddy = l/v ~ Myr (eddy turnover time)
  Re_m ˜ 10¹°    (magnetic Reynolds number)
  Source: BB_C_Equations item 748
```

2.2 Terminal Velocity Buoyancy

```
F_UBii,termv = F_rel × (v(2GM(1-G)/r_launch) / E_LEP) × Q_wave × (t·L/c)

  G = L/L_Edd ˜ 1   (Eddington ratio for wind-driven systems)
  r_launch ˜ 100 R_s  (wind launch radius)
  t = optical depth
  Source: BB_C_Equations item 853, 1823
```

2.3 Hawking Radiation Buoyancy

```
F_UBii,haw = F_rel × (hc³/(8pGMk_B) / E_LEP) × Q_wave × (?/(2p))

  T_H = hc³/(8pGMk_B)  (Hawking temperature)
  ? = c4/(4GM)         (surface gravity)
  Source: BB_C_Equations item 901, 1246
```

2.4 Quasi-Normal Mode Ringdown Buoyancy

```
F_UBii,qnm = -F_rel × (c³/(2pGM) · (0.3737 + 0.088·a_f) / E_LEP)
              × Q_wave × e^{-t/t} × [t ? M]

  a_f ˜ 0.69   (dimensionless BH spin post-merger)
  t = ringdown decay time
  f_QNM = c³/(2pGM_f) · f(0.3737 + 0.088·a_f) (Berti fits l=2, m=2 mode)
  Source: BB_C_Equations item 945, 1293
```

2.5 Blandford-Znajek Jet Power Buoyancy

```
F_UBii,bz = F_rel × (1/32·B²R_H4O_H²/c / E_LEP)
             × Q_wave × (ac/(2R_H)) × (1+t_var)

  R_H = GM/c²  (horizon radius)
  O_H = ac/(2R_H)  (BH angular velocity)
  Source: BB_C_Equations item 1147

F_UBii,bz2 = F_rel × (?/(16p) · F²_BH · O²_BH/c / E_LEP)
```

```
            × Q_wave × (ac³/(2GM)) × 0.05-1

  F_BH = B·p·r_H²  (BH magnetic flux thread; EHT-calibrated)
  Source: BB_C_Equations item 967, 1316
```

2.6 Arnett Supernova Buoyancy

```
F_UBii,arnett = F_rel × (M_Ni·e_Ni/t_d / E_LEP) × Q_wave × e^{-t/t}

  M_Ni = nickel mass (~0.5 M_?)
  e_Ni = nickel decay energy
  t_d² = 3?M/(4pcv²)  (diffusion time)
  Source: BB_C_Equations item 1035
```

2.7 TOV Hydrostatic Equilibrium Buoyancy

```
F_UBii,tov = -F_rel × (dP/dr = -Gm(r)?(r)/r² · (1 + P(r)/(?(r)c²))
            × (1 + 4pr³P(r)/(m(r)c²)) / (1 - 2Gm(r)/(rc²)) / E_LEP) × Q_wave

 Includes all GR corrections (Schwarzschild metric, pressure-energy coupling)
  Source: BB_C_Equations item 1300
```

2.8 Pulsar Spin-Down Buoyancy

```
F_UBii,puls = -F_rel × (t = P/(2?) / E_LEP) × Q_wave

  P = period, ? ˜ 10?¹5 s/s (period derivative, Vela)
  I·dO/dt = -L/O   (torque equation)
  Source: BB_C_Equations item 1302
```

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3. Stellar/Accretion Variants

3.1 Jet Velocity Buoyancy

```
F_UBii,jetvel = F_rel × (v_j ˜ v_K·(r_A/r_0)^{1/2} / E_LEP)
                × Q_wave × (B/v(4p?)) × (1+t/t_A)

  v_K = v(GM/r_0)  (Keplerian velocity at footpoint, r_0 = 1–10 AU)
  r_A = Alfvén radius (10–50 AU, from POETS protostellar data)
  Source: BB_C_Equations item 1096, 1272
```

3.2 Type-I Planet Migration Buoyancy

```
F_UBii,migr = -F_rel × (-f·2(GM_p)² · S/(M_*·O·a5·(H/r)³) / E_LEP)
               × Q_wave × [t = M_p/?_acc ˜ 106 yr]

  f ˜ -1.36  (Lindblad/corotation factor, inward migration)
```

```
  S = disk surface density
  Source: BB_C_Equations item 1121, 1322
```

3.3 Superfluid Vortex Glitch Buoyancy

```
F_UBii,glitch = F_rel × (?? = I_s/I · ?0 · (1-e^{-t/t_q}) / E_LEP)
                × Q_wave × ?O × e^{-t/t_q}

  I_s = superfluid moment of inertia (~10³8 kg·m²)
  t_q = quench timescale
  ?O = angular velocity jump
  Source: BB_C_Equations item 1753, 1304
```

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4. Shock and Blast Wave Variants

4.1 J-Type Shock Buoyancy (Rankine-Hugoniot)

```
F_UBii,jshock = F_rel × ((?1v1² + P1) / E_LEP)
                × Q_wave × (v2/v1) × (?+1)/(?-1+2/M²)

  ? = 5/3  (adiabatic index)
  M = v1/c_s  (Mach number, from Chandra X-rays in HH 154)
  Source: BB_C_Equations item 1193, 1274
```

4.2 Sedov-Taylor Blast Wave Buoyancy

```
F_UBii,sedov = F_rel × ((E·t²/?)^{1/5} / E_LEP) × Q_wave × [d/dt(Mv)=0] × t^{2/5}

  E ˜ 105¹ erg  (explosion energy)
  ? = ambient density
  R(t) = (Et²/?)^{1/5}  (self-similar blast radius)
  Source: BB_C_Equations item 1207, 1288
```

4.3 C-Type Shock Buoyancy (Magnetized)

```
F_UBii,cshock = -F_rel × ((?×B)×B/(4p) + ?_i·?_in·(v_n-v_i) / E_LEP) × Q_wave

 C-shocks: continuous, multi-fluid MHD (ions+neutrals+magnetic field coupled)
  ?_in = ion-neutral collision frequency
  Source: BB_C_Equations item 1276
```

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5. Feedback and Outflow Variants

5.1 AGN Feedback Buoyancy

```
F_UBii,agn = F_rel × (f(v_out)·L_AGN/c / E_LEP) × Q_wave × (?_out·v_out) × v_out?¹
```

```
  Momentum-driven: p_term = ?_out·v_out = f(v_out)·L_AGN/c
  ?_out ˜ 10–100 M_?/yr
  Source: BB_C_Equations item 1165, 1314
```

5.2 Angular Momentum Accretion Buoyancy

```
F_UBii,ang = -F_rel × (?·r²·O / E_LEP) × Q_wave × T_B × e^{-t/t_disk}

  T_B = B²r³/(4p)  (magnetic braking torque)
  t_disk = disk decay timescale
  Source: BB_C_Equations item 1189, 1270
```

5.3 Feedback Energy Coupling Buoyancy

```
F_UBii,coup = F_rel × ((1/2)·?_w·v_w² / (?_acc·c²·10) / E_LEP) × Q_wave × 0.05–
0.1

  e_f = E_kin/(?_acc·c²) ˜ 0.05–0.1  (coupling fraction)
  Source: BB_C_Equations item 1331, 1554
```

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6. Structure Formation Variants

6.1 Press-Schechter Halo Mass Function Buoyancy

```
F_UBii,ps = F_rel × (dn/dM = v(2/p)·(?0/M)·(d_c/s(M))·|d ln s/d ln M|·exp(-
d²_c/(2s²)) / E_LEP)
             × Q_wave × ?O (Gaussian part from s ˜ d_c)

  d_c ˜ 1.69  (critical collapse overdensity)
  Source: BB_C_Equations item 877, 1574
```

6.2 Structure Growth Rate Buoyancy

```
F_UBii,grow = -F_rel × (d̈ + 2H·d? = (3/2)·O_m·H²·d/a³ / E_LEP) × Q_wave

  D(a) = 5O_m/2 · ?0^a da'/(a'H(a')/H0)³  (growth factor)
  Growing mode: D ? a in matter era, suppressed by dark energy
  Source: BB_C_Equations item 1371, 1244
```

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7. GRB Variants

7.1 GRB Fireball Expansion Buoyancy

```
F_UBii,fire = F_rel × (G(r) = r/R0 (r<R_s), G=? (r>R_s) / E_LEP) × Q_wave
```

```
  R0 ˜ 107 cm  (initial fireball radius)
  R_s = ?²R0   (saturation radius)
  Source: BB_C_Equations item 1482, 1306
```

7.2 GRB Afterglow Synchrotron Buoyancy

```
F_UBii,after = -F_rel × (F_? ? ?^{-(p-1)/2}·t^{-3(p-1)/4} (?_m<?<?_c) / E_LEP) × Q_wave

  p ˜ 2.2–2.5  (electron power-law index)
  Electrons accelerated by DSA, spectrum N(E) ? E^{-p}
  Source: BB_C_Equations item 1227, 1308
```

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8. References

- `grok_share_7514fe.txt` lines 2443–2680 (BB_C_Equations_04Sept2025.pdf Part 1 catalogue)
- PAPER_196: Triadic Master Equation System
- PAPER_197: F_U_Bi_i Extended Integral